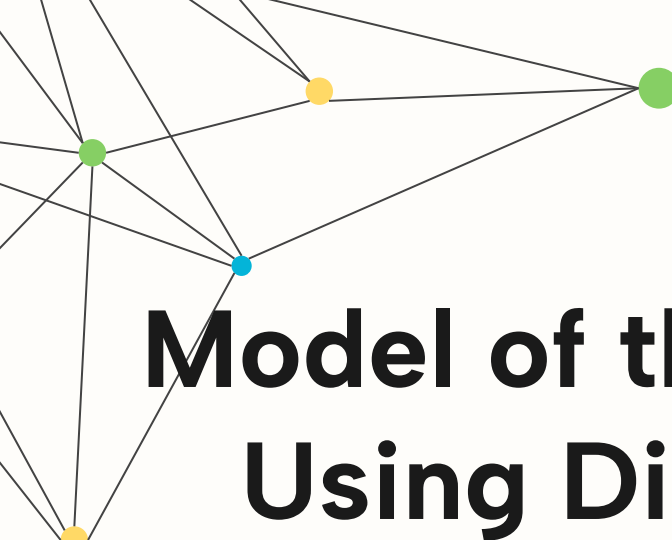
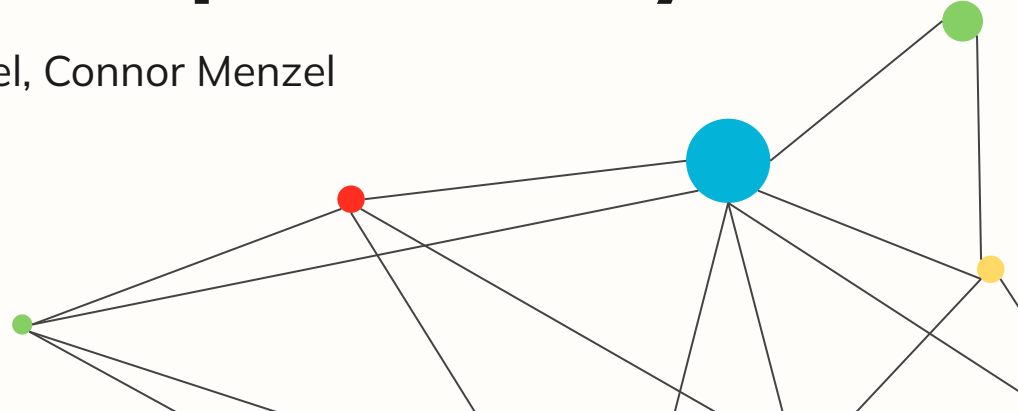


A network graph in the top-left corner with several nodes (green, yellow, blue) connected by thin grey lines.

Model of the Spread of Diseases Using Discrete Time Markov Chains and Graph Theory

A network graph in the bottom-right corner with several nodes (green, yellow, blue) connected by thin grey lines.

Sarah Nayel, Connor Menzel



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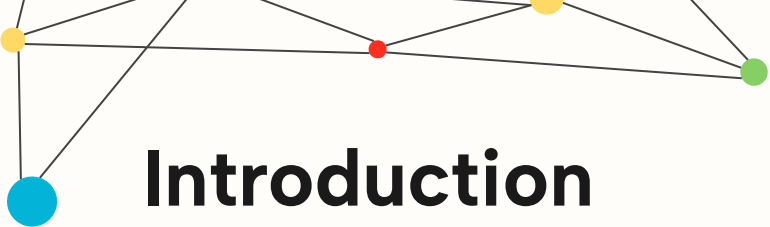
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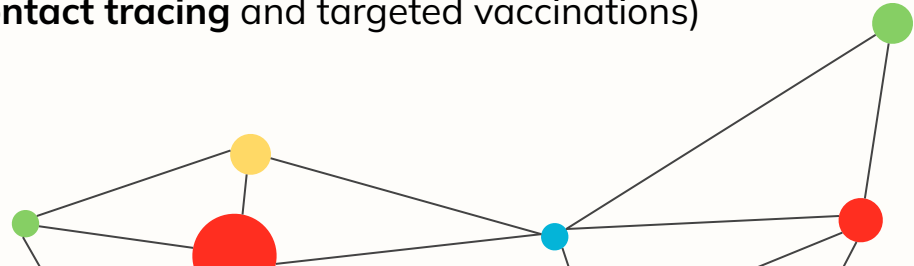




Introduction

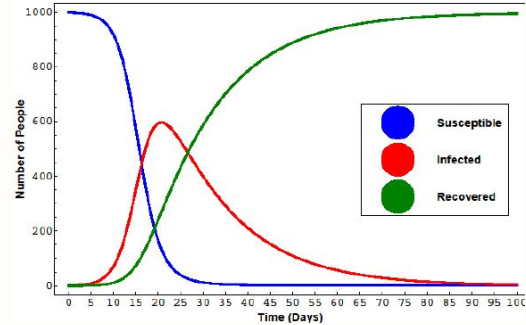


- Infectious diseases spread through contacts between individuals
 - Contacts form a network
- Traditional epidemiology assumes homogeneous mixing
 - I.e. everyone is equally likely to meet everyone
- Network epidemiology involves:
 - Who meets who
 - How frequently
 - **In what structure**
- Applications of understanding a contact network
 - Predict early epidemic growth and estimate the size of the outbreak
 - Improves intervention design (e.g. **contact tracing** and targeted vaccinations)



Motivation

- Goal: Model how diseases spread through populations
- Standard Approach: SIR Model
 - Divides population into Susceptible, Infected, Removed
 - Assumes:
 - Homogeneous mixing (no network/geography)
 - Deterministic dynamics
 - Limitations
 - Ignores who interacts with whom
 - Cannot capture randomness in transmission
- My Question
 - How can we model an epidemic spreading on a network using a stochastic framework?



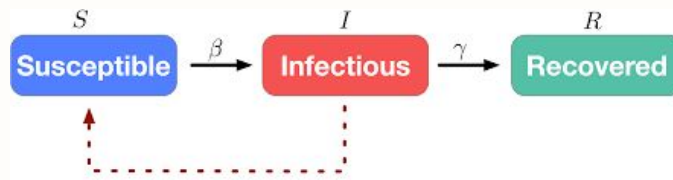
Markov Chains

- **Def:** a type of random process that models a sequence of events
- **Memoryless property:** the state of the next time step depends only on the state in the current time step
 - $P(X_{t+1} | X_t, X_{t-1}, \dots) = P(X_{t+1} | X_t)$
 - “These models rely upon the idea that a system may be modeled as a random walk, where the system transitions from state to state in discrete time-steps (possibly itself again) with a given probability that only depends on the current node of the system” (Kim, 2022)
- Disease spread is inherently random
- Markov chains model this using transition probabilities

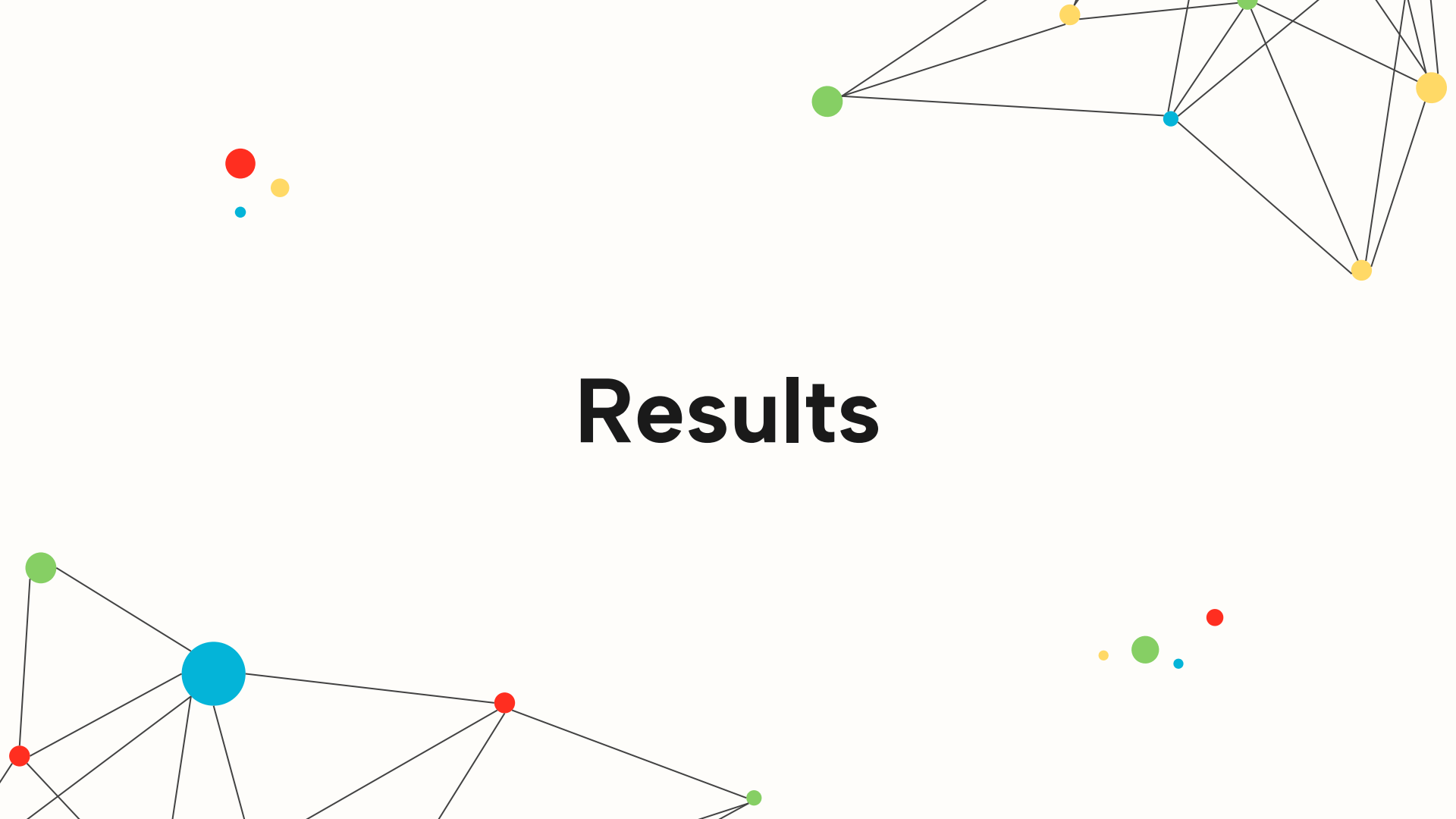
Algorithm

Algorithm Steps (per time step)

- For each node, determine its current state: Susceptible (0), Infected (1), Removed (-1)
- If infected:
 - Recover with probability γ
- If susceptible:
- Count infected neighbors:
 - $n_j = (A \cdot X)_j$
 - Compute infection probability:
 - $p = 1 - (1 - \beta)^{n_j}$
 - Infect with probability p
- Use a random number generator to determine transitions
- Update all node states simultaneously
- Average results over multiple simulation runs for stability

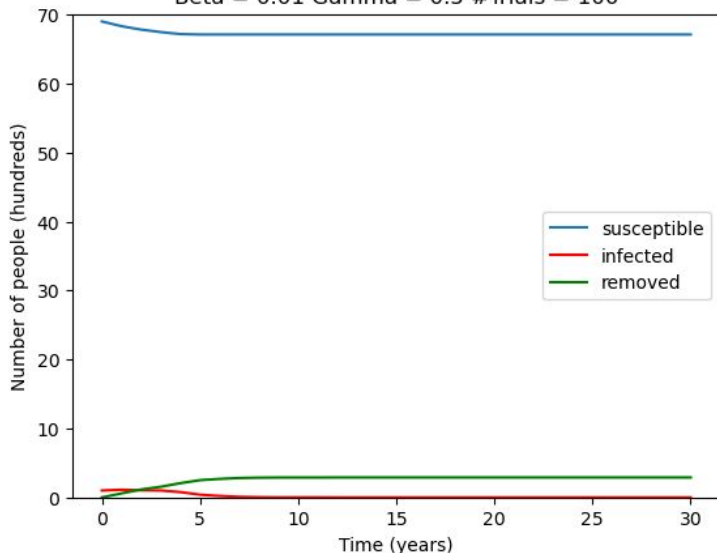


Results



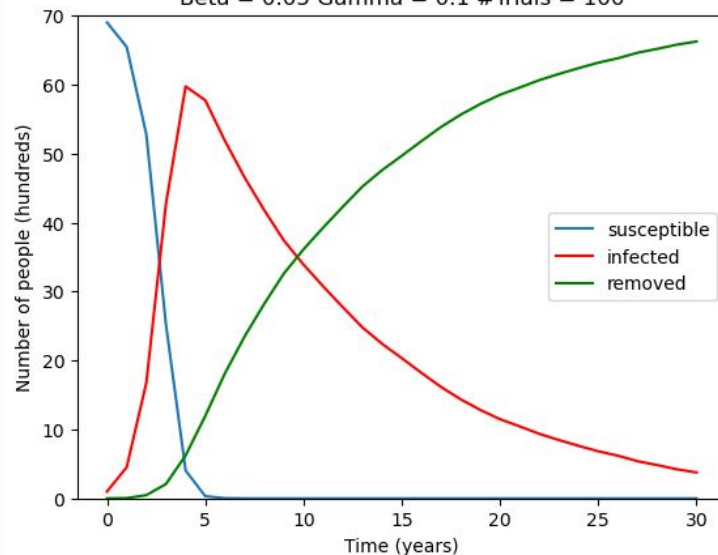
Complete Graph, 1 infected

Averaged SIR Model Simulation of complete graph with 1 infected
Beta = 0.01 Gamma = 0.5 #Trials = 100



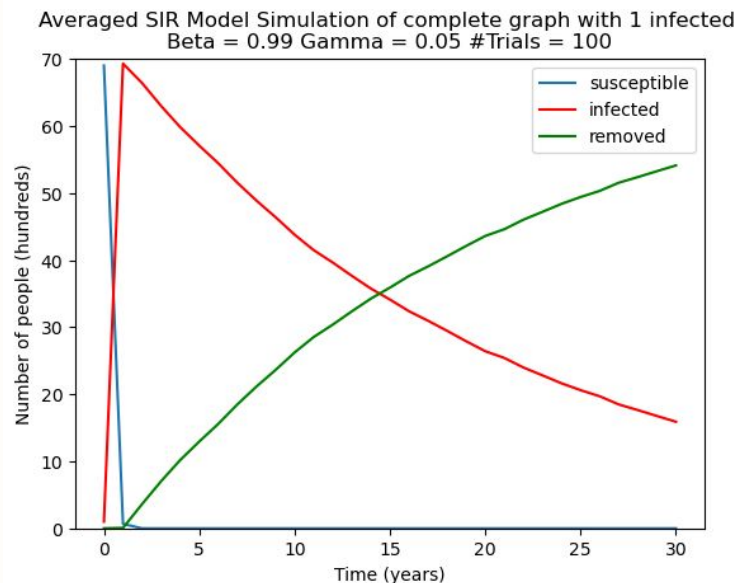
Low infection probability, disease dies out quickly

Averaged SIR Model Simulation of complete graph with 1 infected
Beta = 0.05 Gamma = 0.1 #Trials = 100

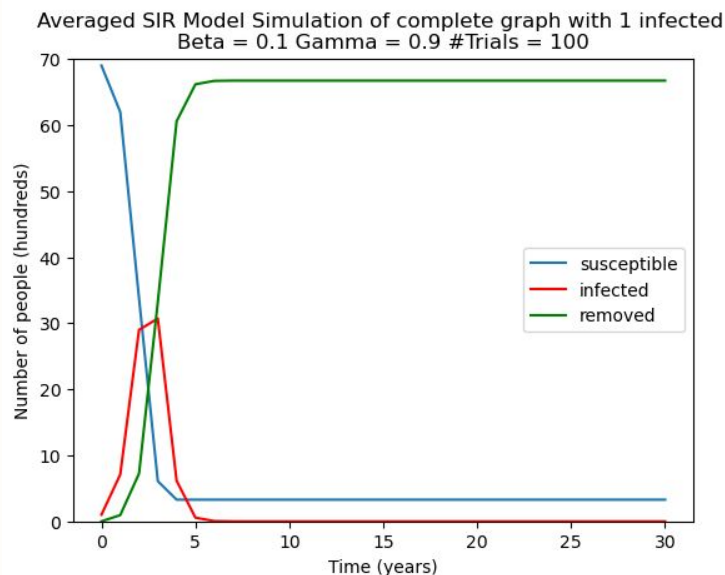


Standard case, closely resembles SIR model

Complete Graph, 1 infected



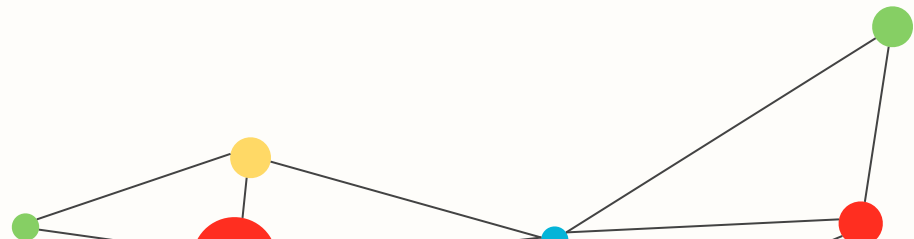
Large outbreak, infected spikes early on



Fast recovery, spread suppressed



Conclusions

- The model is successful and produces graphs that we would expect based on the inputs
 - The stochastic network SIR model produced realistic epidemic behavior
 - Simulation results changed as we expected with:
 - infection rate β
 - recovery rate γ
 - network structure
 - The model captures contact-based transmission and randomness in disease spread
- 



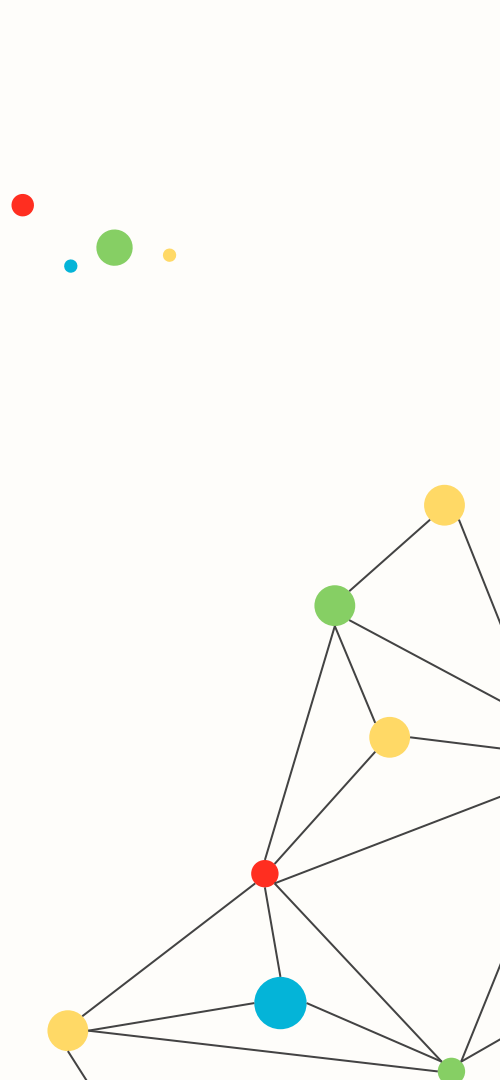
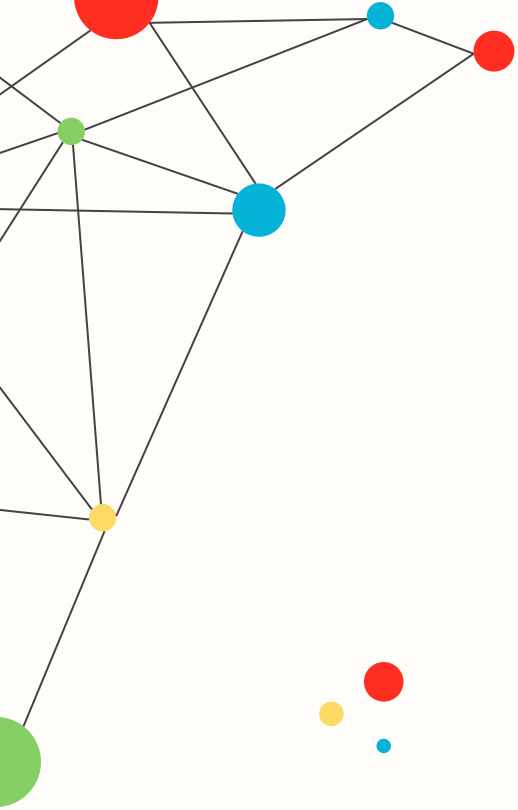
References

Norris JR. 2007. Markov chains. Cambridge: Cambridge Univ. Press.

Kim S, Breen J, Dudkena E. 2022 Jul 7. On the use of Markov chains for epidemic modeling on networks. [accessed 2026 Feb 13]. <https://arxiv.org/pdf/2207.02737v1>.



Danon L, Ford AP, House T. 2010. Networks and the Epidemiology of Infectious Disease. Kerr B, editor. Nihgov. Volume 2011(Article ID 284909):28 pages.
doi:<https://doi.org/10.1155/2011/284909>. [accessed 2026 Feb 13].
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3062985/pdf/IPID2011-284909.pdf>.



Thanks!

Questions?

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